

Arxtron PRJ25106 Robot Tuning Procedure

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Chapter 1

Introduction

1.1 Purpose

This document is intended for users of Arxtron's PRJ25106 Auto Leak Tester station; it outlines the station's robot tuning procedure.

Chapter 2

Tuning Procedure

This procedure should be followed whenever the PRJ25106 station robot needs to be tuned.

2.1 Background

To know where to move, the robot program references a file called **RobotPoints.pts**. This file contains a list of known points, each with its own set of X, Y, Z, and U coordinates. The following tuning procedure aims to set these points such that the robot smoothly picks up and places DUTs when in Auto Mode.

2.2 Setup

Before beginning, ensure that you have a PC with Epson RC+ 7.0 installed.

Additionally, ensure that you have the robot tuning tool, tuning clamp, and a DUT on hand.

2.3 Procedure

1. Return the station to Manual Mode: Using the HMI, put the station in manual mode. Once safe, remove all DUTs from the station.

2. Grip the tuning tool: Move the robot to a position in which the gripper can easily be accessed. Open the gripper and attach the tuning tool to one of the gripper fingers. Close the gripper; the tuning tool should fit snugly.



Figure 2.1 Robot Tuning Tool

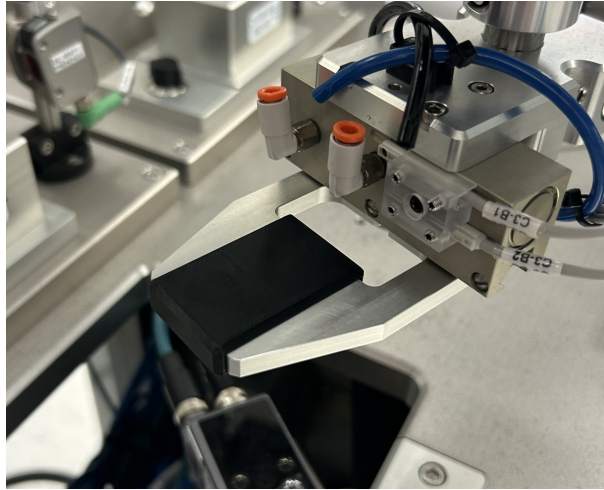


Figure 2.2 Tuning Tool in Gripper

Next, take the tuning clamp, with the flat side facing upwards. Slide it onto the gripper, starting with the wider opening. It should slide all the way to the back of the gripper.

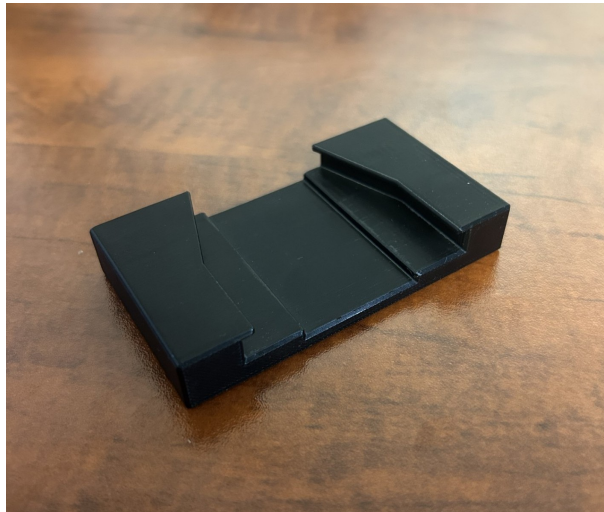


Figure 2.3 Robot Tuning Clamp

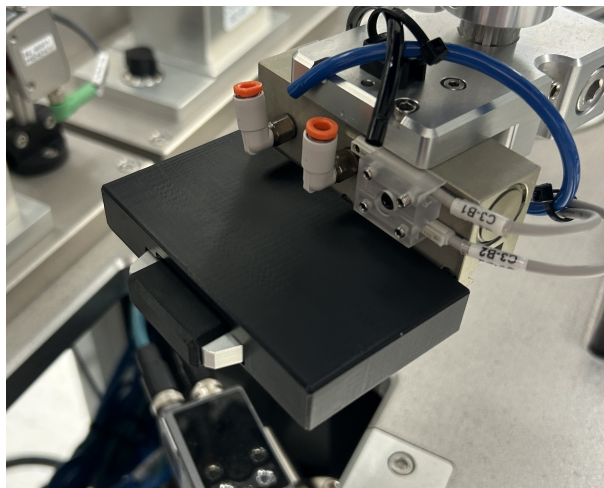


Figure 2.4 Tuning Clamp on Gripper

3. Open Epson RC+: Open the application in program mode, and open the appropriate project file.

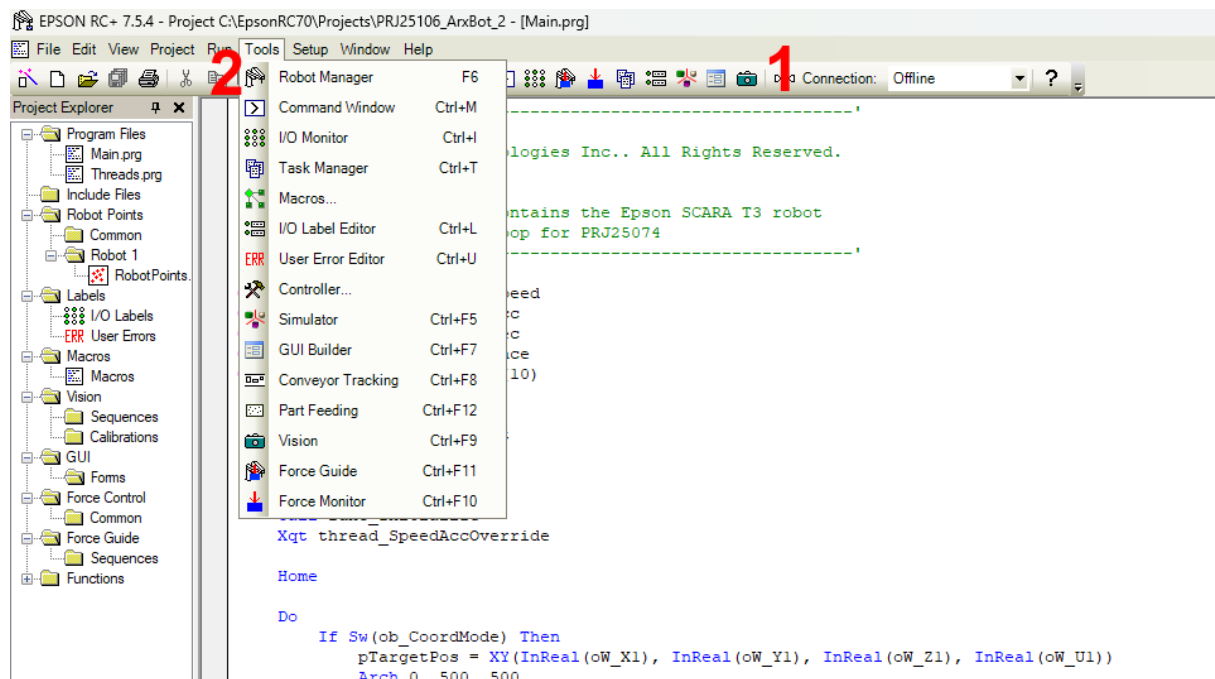


Figure 2.5 Epson RC+ Interface

4. Connect to the robot: First, ensure that your PC is physically connected to the robot, either via USB or Ethernet. Next, navigate to the **Connection** dropdown in RC+ (**Figure 2.5, Element 1**) and connect via the chosen method. Note that this will disable any robot position control from the HMI/PLC.

5. Tune the nest positions: In RC+, navigate to **Tools > Robot Manager > Control Panel** (**Figure 2.5, Element 2**), and select **Free All** to unlock all joint movement. The robot can then manually be moved into the desired position. Once in place, navigate to **Tools > Robot Manager > Jog & Teach**, and under the **Teach Points** tab, select the desired point to write the current position to. Press **Teach** to write to the point. Note that the jogging controls in **Jog & Teach** can be used in the tuning process if desired; the joints must first be locked via **Tools > Robot Manager > Lock All**.

For nest positions, it is recommended to manually move the robot such that the tuning tool sits in the nest similar to how a DUT would. Once in place, give it a light shake to help it settle into the optimal position.

6. Tune non-nest positions: Following a similar process to the previous step, manually position the robot and save each point to tune the pass, fail, and barcode positions. Note that the barcode is easier to tune while directly monitoring the barcode scanner (see [Barcode Scanner Monitoring](#)).

7. Adjust point Z-values: The tuning tool and clamp are useful for setting X,Y, and U coordinates. However, the robot points may need to be translated along the Z-axis in order to properly grip the necessary DUTs. This can easily be done by jogging the robot to a nest, placing a DUT in the nest for reference, and jogging the robot's Z-coordinate until a satisfactory position is reached. Note the difference in Z-position between the current position and the original point, and manually apply the offset to all points from the points display. This can be done from the points display (**Tools > Robot Manager > Points**).

Number	Label	X	Y	Z	U
▶ 0	pLoadingNest	43.777	387.347	-42.775	167.281
1	pBarcode	149.652	115.599	-29.669	57.991
2	pTestingNest1	290.183	175.551	-76.340	1.458
3	pTestingNest2	290.783	-74.756	-76.347	1.745
4	pPassChute	181.053	-110.092	-44.753	-65.152
5	pFailChute	213.125	84.071	-131.113	-88.375
6					

Figure 2.6 Robot Points Display

8. Send points to the controller: After manually adjusting any point values within the points display, press the **Save** button at the bottom of the screen, which will send the points to the controller.

9. Disconnect from the robot: Once the points have been sent to the controller, the connection to the robot can be terminated. The robot will switch into run mode, re-homing automatically. Before running the station again at full speed, it is recommended that the newly-tuned positions are tested in manual mode at a low speed (5-10%).

Chapter 3

Notes

3.1 Home Position

The robot's home position is set within the code itself (**Figure 3.1, Element 1**). Typically, the home position should be set to the same coordinates as the **barcode** position. Rebuild the program after making changes (**Project > Rebuild**).

3.2 XYZ Limits

The robot's XYZ limits are also set within the code (**Figure 3.1, Element 2**). If a point contains a coordinate that is out of range, you may adjust the range (assuming it is safe to do so). Rebuild the program after making changes.

```
Function func_Initialize
  On ib_Initializing

  Call func_ResetOutputs

  Reset          'Reset errors, in case ESTOP was activated
  Motor On       'Turn the motor on
  Power High     'Set power to high

  If ErrorOn Or EStopOn Or Motor = 0 Or Power = 0 Then
    On ib_Error
    Pause
  EndIf

  ' Torque limit copied from customer
  TCLim 30, 30, -1, 10 'Set TC Lim Values

  LimZ 0          'Default jump ceiling

  ' Home values obtained from manually positioning the robot
  HomeSet 197394, -285096, 0, 19874
  Hordr &B0100, &B0011, &B1000, &B0000

  ' Limits set based on tuned positions, may need to change
  XYLim 40, 297, -113.0, 390.0, -133, 1

  ' Values copied from customer
  alphaDefaultSpeed = 75
```

Figure 3.1 Robot Code Snippet: func_Initialize

3.3 Barcode Scanner Monitoring

AutoID Network Navigator can be used to monitor the barcode scanner. Ensure that your PC is physically connected to the scanner, then launch the application. The application will automatically detect the scanner; simply follow the prompts to connect and press the **Monitor** button once ready.

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